

Diagnosis and treatment of pyo-granulomatous panniculitis due to *Mycobacterium smegmatis* in cats

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Journal of Small Animal Practice (1994) **35**, 524-530

ABSTRACT

Four cats with extensive pyogranulomatous panniculitis and dermatitis due to *Mycobacterium smegmatis* were evaluated over an eight-month period. All were treated initially with doxycycline (50 mg per cat every eight to 12 hours) for several weeks, then subjected to radical surgical excision of infected tissues. Reconstructive surgery utilising advancement flaps was required to close wounds without undue tension. Postoperatively, the cats received parenteral gentamicin (2 mg/kg every eight hours) for three to five days, followed by orally administered ciprofloxacin (62.5 mg per cat every eight hours or 125 mg per cat every 12 hours) for three to six months. Infection was eradicated in all the cats, with no signs of recurrence 15 to 23 months after surgery.

INTRODUCTION

Mycobacteria are Gram-positive rods which have as their distinguishing feature a cell wall rich in high molecular weight fatty acids and waxes. This imparts two characteristic features to the genus – a capacity for survival and replication within phagocytes, and the ability to retain carbolfuchsin stain after heating and exposure to acid and, or, alcohol (Youmans 1980). Mycobacteria of clinical importance include the tubercle bacillus *Mycobacterium tuberculosis* which causes tuberculosis in man and cats (Wilkinson and Mason 1991), the obligate pathogen *M leprae-murium* which causes 'rat leprosy' and nodular cutaneous disease in cats (Schiefer and Middleton 1983), and a saprophytic rapidly growing group (formerly referred to as Runyon's [1959] group 4) that are ubiquitous in the environment and behave infrequently as opportunistic pathogens in man and cats (Grange and Yates 1986, Pedersen 1988, Wilkinson and Mason 1991). Bacteria in this latter group have been reported to produce cutaneous and subcutaneous disease in cats and include *M fortuitum*, *M chelonae* (formerly *M chelonae*), *M thermoresistible* and *M smegmatis* (Dewevre and others 1977, Wilkinson and others 1978, 1982, Kunkle and others 1983, White and others 1983, Willemse and others 1985; for a review see Pedersen [1988] or Wilkinson and Mason [1991]). Wilkinson and colleagues believe *M smegmatis* infections are distinctive because infection principally involves the subcutaneous fat, whereas infection with the other saprophytic mycobacteria initially affects both the dermis and the subcutis (Wilkinson and others 1982, Wilkinson and Mason 1991).

M smegmatis has a wide distribution in nature, its habitats include soil (Tsukamura 1976) and free-standing bodies of water (Beerwerth 1973). It was first reported as a cause of pyogranulomatous dermatopanniculitis in cats by Wilkinson and others (1982). Infection was thought to follow the traumatic introduction of organisms into subcutaneous tissues through penetrating skin wounds. Such wounds may occur during cat fights, either as a result of bites, or deep claw wounds in which the victim's ventral abdomen is raked by its opponent's hindclaws (Wilkinson and Mason 1991). Penetrating injury by sticks, vegetable matter and vehicular trauma could also give rise to infection (Wilkinson and others 1982, Newton and others 1993).

M smegmatis infections are more common in cats with substantial subcutaneous fat deposits, and the inguinal fat pad is the most commonly affected site (Wilkinson and others 1982, Kunkle and others 1983, White and others 1983, Studert and Hughes 1992, White and Kowalski

1992). Experimental and clinical observations suggest that adipose tissue offers a favourable environment for the survival and proliferation of saprophytic mycobacteria, either by providing triglycerides for growth of the organisms, or protecting them from the phagocytic or immune response of the host (Hagan and Levine 1932, Richardson 1971, Wilkinson and others 1982). Initial reports suggested warm humid climates favoured infection (Wilkinson and Mason 1991), but recent cases from a variety of regions indicate that this is not necessarily so (Studdert and Hughes 1992, White and Kowalski 1992).

Infection gives rise to a distinctive clinical entity. Often there is a history of recent trauma to the inguinal, perineal or axillary area, usually as a result of a cat fight. If the animal is presented promptly for veterinary attention, a circumscribed plaque or nodule may be present at the site of injury. Later there is progressive thickening of the nearby subcutis to which the overlying skin becomes adherent (Figs 1A and 2A). Affected areas become denuded of hair and numerous punctate fistulae appear, discharging a watery exudate (Figs 1B and 2A). Fistulae are intermingled with focal purple depressions (Fig 1B) which correspond to thinning of the epidermis over accumulations of pus. The lesion gradually increases in area and depth, and may eventually involve the entire ventral abdomen, adjacent flanks or limbs. When the entire inguinal panniculus is involved the area becomes board-like, up to 3 cm thick and firmly attached to the abdominal wall. Lesions of this type are virtually pathognomonic for opportunistic mycobacterial infection (Monroe and others 1988, Kunkle 1993), and it is the opinion of Wilkinson and others (1982) and Wilkinson and Mason (1991) that they are characteristic of disease due to *M. smegmatis*. When cats are presented for veterinary attention early in the course of the disease, the lesion may be confused with an anaerobic cat bite abscess if laboratory studies are not undertaken. Standard treatment consisting of surgical drainage and administration of a synthetic penicillin is typically followed by wound breakdown and development of a large non-healing sinus tract surrounded by indurated suppurating granulation tissue (Monroe and others 1988, Kunkle 1993).

Histologically, there is pyogranulomatous inflammation of subcutaneous adipose tissue, overlying dermis and underlying abdominal fascia and musculature. Some severely affected cats become depressed, pyrexia, inappetent, lose weight and become reluctant to move. Surprisingly, other cats remain well constitutionally despite extensive pyogranulomatous disease.

Formerly, advanced infections due to *M. smegmatis* and related organisms were considered

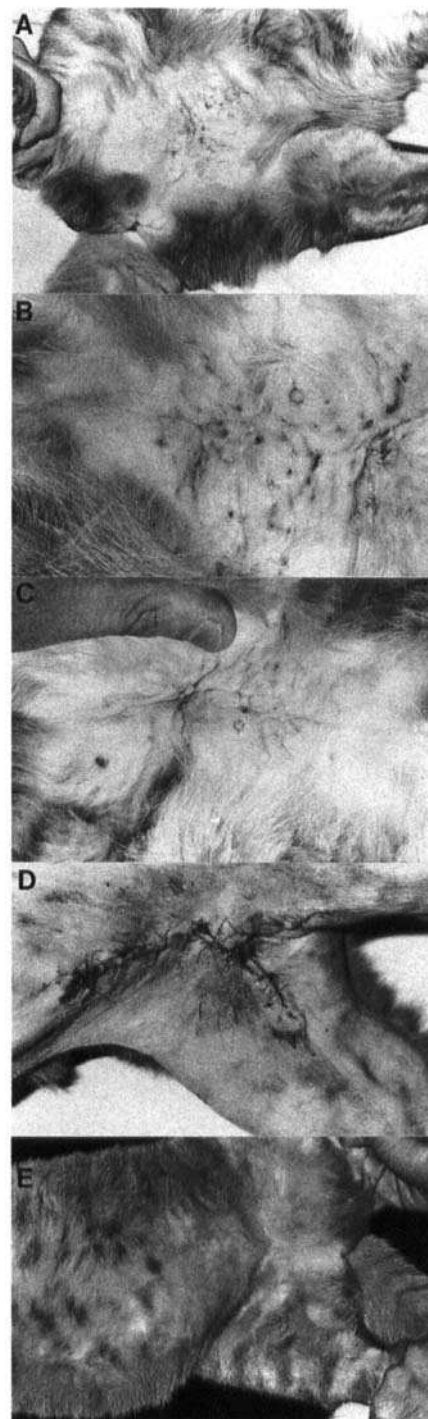


FIG 1. *Mycobacterium smegmatis* infection of the inguinal fat pad and overlying skin of case 1. The inguinal region before therapy is shown in (A). A close-up (B) shows numerous draining sinus tracts. (C) shows the improved appearance of the lesion following four weeks of orally administered doxycycline, immediately before surgery. The appearance of the inguinal region at suture removal and six months after surgery are shown in (D) and (E), respectively

difficult or impossible to cure (Wilkinson and others 1982, Kunkle and others 1983), although some cats responded to repeated aggressive surgical interventions and combinations of antimicrobials (White and others 1983, Wilkinson and Mason 1991). Recent reports emphasise the value of fluoroquinolone antibiotics administered systemically and, or, topically (using

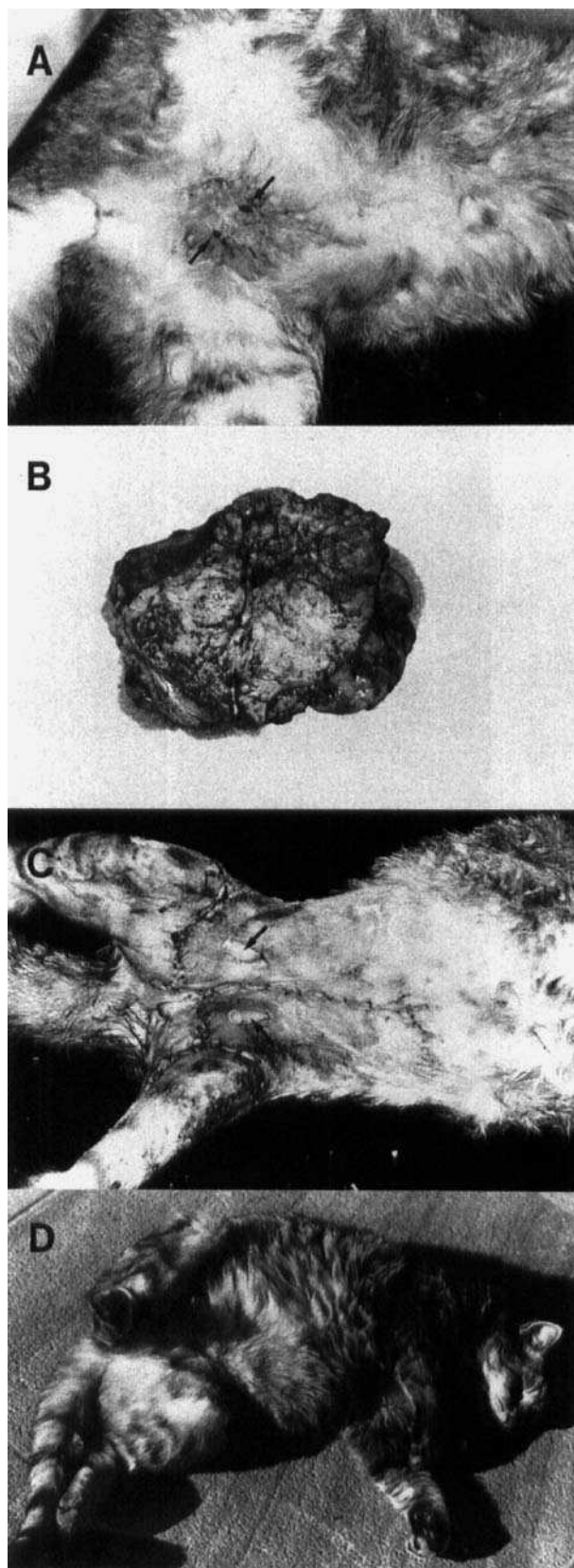


FIG 2. *Mycobacterium smegmatis* infection of the inguinal fat pad and overlying skin of case 4. The inguinal region before therapy is shown in (A); arrows point to draining sinus tracts. Infected tissues removed by en bloc resection are displayed in (B), skin-side facing up. (C) shows the surgical wound which was closed by advancing skin flaps mobilised from the flank folds of the pelvic limbs. Arrows highlight Penrose drains exiting the subcutaneous space through separate stab incisions. The appearance of the inguinal region six months following surgery is shown in (D)

dimethylsulphoxide) to treat these infections (Studdert and Hughes 1992, White and Kowalski 1992). Given the extent and severity of the pathology, and the propensity for organisms to survive and replicate within macrophages, basic principles suggest that the best chance for a long-term cure resides with aggressive surgical resection followed by adequate drainage and long courses of appropriate antibiotics. The present report concerns the authors' experience with four consecutive *M smegmatis* infections treated successfully in this manner.

CASE HISTORIES

Four cats with *M smegmatis* infections were diagnosed at the authors' hospital between March and November 1992. All were female domestic shorthaired cats referred for evaluation of refractory skin disease. They ranged in age from 18 months to eight years and came from diverse geographical regions of Sydney and its environs. Cats were considered obese by their owners before infection, and two owners believed the problem started following fight injuries. Clinical and microbiological data from these cases are summarised in Table 1.

The inguinal fat pad, overlying skin and adjacent tissues were affected in three cases (Figs 1A and 2A); the remaining infection was centred in the subcutaneous tissues of the cat's right axilla. In all the cases, infection had spread extensively to involve contiguous cutaneous and subcutaneous tissues, including the muscle and fascia of the abdominal wall in two cats. Infections were longstanding, the problem starting four to nine months before admission. Three of the cats were either pyrexial (case 2, 40.2°C and case 3, 39.6°C), inappetent or had lost weight, while the remaining cat was systemically well. The cat with axillary involvement was lame and the affected leg was painful on manipulation.

All the cats were managed similarly. A tentative diagnosis of mycobacteriosis was confirmed by collection under general anaesthesia of pus and deep tissue samples for microbiology and histopathology. Biopsies were triturated in brain heart infusion broth using a sterile mortar and pestle. Smears prepared from the resulting homogenate and swabs of the purulent exudate were stained using Burke's modification of the Gram stain and a modified acid-fast procedure (decolourising with 5 per cent sulphuric acid for three minutes).

In all the cases it was possible to visualise Gram-positive and, or, acid-fast rods in smears, although an exhaustive search of several smears was sometimes required. Many organisms showed beading. Tissue homogenates and pus were streaked on to duplicate 5 per cent sheep

Table 1. Clinical and microbiological data pertaining to four cats with *Mycobacterium smegmatis* infections

Case	Age (years)	Sex	History of cat fight	Location of lesion	Duration of signs (months)	Infection acquired (season)	Systemic signs	Acid-fast organisms Smears	Tissue sections	Culture
1	8	FN	—	Inguinal and proximal hindleg	9	November (Spring)	+	+	+	Heavy growth
2	1.5	FN	+	Right axilla and ventral chest	4	May (Autumn)	+	+	+	Heavy growth
3	4	FN	+	Inguinal and ventral abdomen	5	October (Spring)	+	+	—	Heavy growth
4	7	FN	—	Inguinal	9	March (Autumn)	—	+	ND	Moderate growth

FN Neutered female, + Present, — Absent, ND Not done

blood agar plates and 1 per cent Ogawa egg yolk medium (Ogawa and Motomura 1970) and incubated aerobically. Moderate to heavy growth of pin-point, smooth, non-haemolytic colonies was first visible after 36 to 48 hours on sheep blood agar at 37°C. If contamination precluded the primary isolation of acid-fast organisms, tissue homogenates were treated with 4 per cent sodium hydroxide followed by neutralisation with dilute hydrochloric acid before inoculation on to media. In one instance, rapidly growing mycobacteria were selectively differentiated from contaminant skin flora by primary isolation around antibiotic discs (first generation cephalosporins, isoxazolyl penicillins, lincosamides and macrolides) on sensitivity agar. Disc diffusion sensitivity tests were based on the methods of Wallace and others (1979, 1985). Suspensions of each organism in saline or nutrient broth were sown on to Oxoid Iso-Sensitest agar (Oxoid; Basingstoke) and incubated at 37°C. The results were recorded at 48 and 72 hours. All *M. smegmatis* isolates displayed similar sensitivity patterns, with consistent in vitro susceptibility to tetracyclines, gentamicin and the fluoroquinolones. Isolates were sent to a Mycobacteria Reference Laboratory (Mr D. Dawson, Laboratory of Microbiology and Pathology [Mycobacteria Section], State Health Laboratory, Brisbane) for species identification.

Routine histological sections of biopsies stained with haematoxylin and eosin and Putt's modification of the Ziehl-Neelsen method (using 0.5 per cent hydrochloric acid in 70 per cent alcohol) were examined using conventional light microscopy. Pyogranulomatous inflammation was present in all the samples, but acid-fast rods could only be detected in two of three biopsies available for examination, even after a long

search of the sections. As reported by Wilkinson and others (1982) and Kunkle and others (1983), acid-fast bacilli were mostly detected in cleared spaces, presumably extracellular lipid vacuoles. Saprophytic mycobacteria are relatively impermeable to basic dyes because of their lipid-rich cell walls (Wolinsky 1973), and are generally only weakly acid-alcohol fast. Thus only a small proportion of organisms present in the sections are stained successfully, while leaching of lipid vacuoles during processing further reduces the number of detectable bacilli.

While awaiting laboratory results, antimicrobial therapy was initiated using high doses of doxycycline (50 mg per cat every eight to 12 hours), which were continued for two to eight weeks. Cats that were systemically unwell improved with doxycycline therapy; fevers resolved, appetites improved and the cats gained weight. There was a corresponding improvement in the appearance of the infected tissues, with healing of many draining sinus tracts and drying of the overlying skin (Fig 1C). But although the surface and periphery of the lesion improved, the board-like feel of the subcutaneous tissues was unchanged after four, eight, three and two weeks of doxycycline therapy (cases 1 to 4, respectively). Case 3 subsequently received ciprofloxacin (62.5 mg every 12 hours) for 17 days without further benefit.

Surgery was performed in each case after preliminary doxycycline therapy. After anaesthetic induction, an indwelling cephalic catheter was placed for intra- and postoperative fluid therapy, and gentamicin (4 mg/kg) was administered intravenously. Radical surgical excision of infected tissues was performed using sharp and blunt dissection. Haemostasis was achieved using electrocautery. Involvement of the subcutaneous

tissues was invariably more extensive than suggested by the appearance of the overlying skin, an observation which has been made previously in affected human patients (Plaus and Hermann 1991). In three of four cases it was possible to excise all visibly affected tissues. In the remaining cat (case 3) there was involvement of the full thickness of the ventral abdominal wall, of which only the superficial portion was resected. Penrose drains were placed in the subcutaneous space and exited through separate stab incisions. The subcutis was closed using synthetic absorbable sutures (3/0 polyglactin or polydioxanone). Wound closure was challenging in all cases. In case 1 (Fig 1D) and case 3, closure was achieved using relaxing incisions and advancement flaps, while in two other cats it was necessary to mobilise adjacent areas of unaffected integument, namely the elbow folds (case 2) and flank folds (case 4; Fig 2C) (Hunt 1995). Careful apposition of remaining subcutaneous tissues was undertaken to minimise tension on the skin incisions, which were closed using interrupted sutures of 3/0 polypropylene in a cruciate or horizontal mattress pattern. Recovery in the three cats subjected to extensive dissection was painful, and each required morphine analgesia for 12 hours postoperatively.

Gentamicin was administered intravenously or subcutaneously (2 mg/kg every eight hours; Jernigan and others 1988) for three to five days following surgery. During this period, the cats were maintained on intravenous fluids at approximately 150 per cent of maintenance requirements (4 ml/kg/hour) to guard against aminoglycoside-induced nephrotoxicity (Brown and Barsanti 1986). Medication was then altered to orally administered ciprofloxacin (125 mg every 12 hours in cases 1, 2 and 4; 62.5 mg every eight hours in case 3; Watson 1991). The wounds discharged sanguineous fluid for several days following surgery, after which time the drains were removed. The bandages were changed daily during this period except in those cats whose personality dictated less frequent interventions. The cats were discharged from hospital seven to 10 days following surgery. Skin sutures were left in place for at least three weeks, until wound healing was well advanced. Orally administered ciprofloxacin was continued for four, four, six or three months (cases 1 to 4, respectively), although in cases 2, 3 and 4 the dose was reduced by half after three, three or one month, respectively, because of the high cost of medication. Wounds healed uneventfully in all the cases, and all the cats, even those that were not previously ill, gained weight following surgery. Recurrent mycobacterial disease has not been observed at the time of writing, 15 to 23 months following surgery (Figs 1E and 2D).

DISCUSSION

Despite longstanding and extensive *M smegmatis* infection, all the cats were treated successfully using a combination of radical surgery and long-term antimicrobial therapy. Aggressive debulking of infected tissues was considered the cornerstone of effective therapy. The complicated antibiotic regimen evolved for pragmatic reasons during management of these cases. Doxycycline was chosen for initial therapy because of its low cost, and because all *M smegmatis* isolates so far encountered by this laboratory have been sensitive to the tetracyclines in vitro (D. I. Wigney, unpublished observations). Enrofloxacin is not available for veterinary use in Australia, while the two fluoroquinolones marketed for human use (ciprofloxacin and norfloxacin) are expensive. Accordingly, the authors reserved the use of ciprofloxacin for the post surgical period when it was likely to be most beneficial and cost-effective. Gentamicin was administered intraoperatively and in the early postoperative period because it was bactericidal, available in a parenteral form, inexpensive and displayed good in vitro activity against *M smegmatis* isolates. Apart from assiduous attention to wound toilet, the postoperative management of these animals was routine.

The post surgical course of ciprofloxacin was important to ensure primary intention wound healing and prevent recurrent infection. Fluoroquinolones are currently the drugs of choice for most rapidly growing mycobacterial infections because they are bactericidal, penetrate well into tissues including fat, and are concentrated in polymorphs and macrophages (Leysen and others 1989, Hooper and Wolfson 1991). Ciprofloxacin was chosen rather than norfloxacin because of its superior pharmacokinetics and tissue disposition (Leysen and others 1989, Walker and others 1989, 1990, Hooper and Wolfson 1991). Enrofloxacin would be expected to be similarly effective (Studdert and Hughes 1992), and less expensive where available. The optimal duration of post surgical antimicrobial therapy has not been determined, and the present authors are uncertain as to whether the long courses administered were necessary. However, apart from diarrhoea in one cat, side effects were not observed during long-term ciprofloxacin therapy.

A recent report recommends the use of orally administered enrofloxacin without surgery for treating deep-seated mycobacterial infections (Studdert and Hughes 1992). Although clinical cures were reported using this approach, others have been less successful (White and Kowalski 1992, D. Lewis, personal communication). Given the extent and severity of the pathology in many

of these cats, it seems optimistic to expect adequate levels of any antimicrobial to be achieved throughout all involved tissues. The present authors believe that the best chance for a successful long-term outcome is to excise as much infected tissue as possible. Residual foci of infection can then be targeted by the higher concentrations of antibiotics achieved following surgery. Similar recommendations have been made for humans infected with *M. smegmatis*, where antibiotic therapy is considered to be adjunctive to aggressive surgical debridement (Wallace and others 1988, Plaus and Hermann 1991, Newton and others 1993).

Preliminary medical treatment of severely affected cats is indicated to improve their overall status before surgery, to decrease the quantity of infected tissues to be excised, and to reduce the possibility of wound dehiscence. Doxycycline, ciprofloxacin or enrofloxacin are all probably suitable for such pre-operative therapy, with the choice depending on cost and availability.

A high index of suspicion for mycobacterial infections is warranted in 'cat fight abscesses' which fail to heal despite conventional therapy, and when characteristic lesions develop near the inguinal fat pad. Although considerable surgical expertise was required to deal with the advanced cases described in the present report, less radical procedures would probably suffice early in the course of infection if followed by appropriate antimicrobial therapy. In such instances, the authors recommend therapy with either doxycycline or a fluoroquinolone while awaiting histopathology and microbiology results. Recently, the authors consulted with a colleague concerning a cat from which *M. smegmatis* was cultured from a circumscribed pyogranulomatous nodule in the inguinal fat pad. The lesion, which was excised completely and found to contain a wood splinter, did not recur following surgery and a two-week course of ciprofloxacin.

ADDENDUM

Since submitting this manuscript, the authors have successfully managed a fifth case using an identical protocol. The cat, a four-year-old castrated British blue, presented in September 1993 with characteristic involvement of the subcutis and skin over the ventral chest and abdomen of six months' duration. The cat is currently doing well 10 months following surgery, with no sign of disease recurrence.

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BOOK REVIEW

Update on feline medicine

Consultations in Feline Internal Medicine. By J. R. August. Published by W. B. Saunders, Philadelphia. £58.00. Hardback. 680 pages. 1994.

THIS book is not a definitive text on feline internal medicine but an update on various aspects of internal medicine provided by some 108 contributors. The layout is similar to that used in 'Current Veterinary Therapy' and comprises a series of 10 sections (80 chapters) on the following subjects: Infectious diseases, gastrointestinal system, endocrine and metabolic diseases, dermatology, cardiology and respiratory disorders, urinary system, neurology, haematopoietic and lymphatic system, oncology and population medicine.

Each chapter is well laid out with clear headings and subheadings making it easy for the clinician to read or scan the text for information. There are numerous tables and figures throughout to illustrate points or summarise information. I was disappointed at the lack of colour for the endoscopic photographs but suspect this was done in order to reduce costs.

The chapters are produced to a high standard and provide the clinician with up-to-date information on many aspects of feline medicine. It would be wrong to select individual chapters as they are all of a high standard. However, I found the chapters on advances in feline immunodeficiency virus, feline leukaemia virus and feline infectious peritonitis of particular value as was the section on endocrinology which provided

useful information on the diagnosis and treatment of thyroid disease and diabetes mellitus. The section on feline oncology offers the clinician practical information and thus the confidence to tackle this difficult area of feline medicine. The management of lymphoma, soft tissue sarcomas and mast cell tumours are covered together with a chapter on the clinical application of radiotherapy. The chapters on population medicine provided information on the epidemiology and control of various diseases in catteries together with advice on selection of breeding stock to avoid genetic disorders. The final chapter deals with feline nutrition, including nutritional requirements, life-stage feeding and dietary management of a small selection of conditions including feline lower urinary tract disease, obesity and hepatic lipidosis. It is unfortunate this latter part had not been expanded. An earlier chapter discussed enteral and parenteral nutrition but only in relation to gastrointestinal disease.

In summary I feel the editor is to be congratulated in putting together contributions from 108 authors in a manner which is both readable and informative. The book is a must for anyone routinely involved in feline medicine and who wishes to keep up-to-date with the latest information. It should not be considered a definitive text on feline medicine, and where this is required the clinician should look elsewhere. At £58.00 I believe the book represents very good value for money.

J. W. SIMPSON